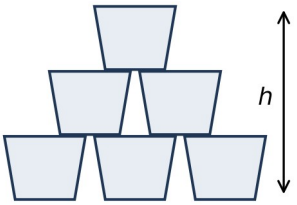
Algebra 1 – Lab 3: Cup Pyramid → Functions Name: _____ Period: _____

Module 3 — Relations and Functions Materials: 15 cups per team · tape measure (cm) · pencil

Team roles: 2 builders · measurer · recorder · grapher · checker (counts every cup and checks every number). Build on the floor or a big table. Cups stand UPRIGHT; each cup rests on the rims of the two below it.

Part 1 — Build it, measure it. Build one row at a time. After EACH row, count every cup and measure the height from the table to the top rim.



Rows, r	Cups in the bottom row	Total cups, c	Height, h (cm)
1			
2			
3			
4			
5			

One cup is _____ cm tall.

★ **Think deeper:** BEFORE you build row 6, predict both its total cups and its height. Then decide: do you have enough cups to check?

predict: c = _____ h = _____

a. Go down each column. By how much does it change each time you add a row?

Height changes by:

_____

Total cups change by:

_____

b. Which column grows by the SAME amount every row? Why does the pyramid force that?

_____

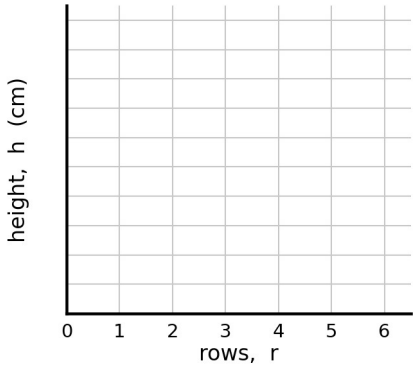
_____

★ **Think deeper:** Why does the cups column NOT grow the same way? Look at the bottom row.

_____

Part 2 — Two graphs. Graph your table. Choose and label a scale on the vertical axis so your points fill the grid.

a. Height vs. rows

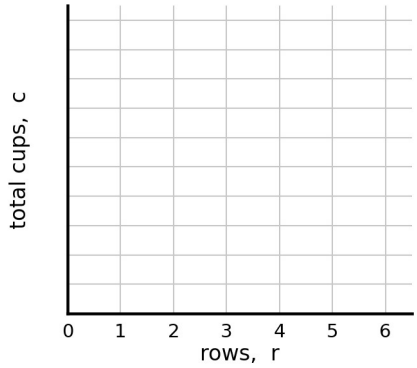


Same change each step ⇒ points fall on a line

★ **Think deeper:** Where would the point for r = 0 be? Does that point make sense for a pyramid?

_____

b. Total cups vs. rows



Changing jumps ⇒ points curve

★ **Think deeper:** The jumps were +2, +3, +4, +5. Predict the next two jumps, then the total cups for 7 rows.

_____

c. Answer for BOTH graphs.

Is each one a function? How do you know?

_____

_____

Linear or nonlinear?

_____

Discrete or continuous? Should you connect the dots?

_____

_____

★ **Think deeper:** Turn it around: is rows a function of height? Is height a function of total cups?

_____

Algebra 1 – Lab 3: Cup Pyramid → Functions Name: _____ Period: _____

Module 3 — Relations and Functions

Part 3 — Rules, domain, and range. Use YOUR cup height. Every answer should mean something about the pyramid.

a. Write a rule for the height.

h = _____ · r

What does the number in front of r mean in the pyramid?

_____

_____

What is the h-intercept? What does it mean?

_____

★ **Think deeper:** Rex used his 9-cm cups. Write his rule. At how many rows is your pyramid 30 cm taller than his?

_____

_____

b. Your team has 15 cups.

Domain of the height function:

_____

Range:

_____

Domain = the r-values you can BUILD. Range = the h-values you get.

★ **Think deeper:** Why is r = 6 not in your domain, even though h = 6 · 12 makes sense to calculate?

_____

_____

c. The total cups follow this rule:

c = r(r + 1) ÷ 2

Check it against your table for r = 4 and r = 5.

_____

_____

How many cups does a 10-row pyramid need?

_____

★ **Think deeper:** Ky says: "The cups graph goes through (0, 0), just like the height graph, so it must be linear." Is Ky right?

_____

_____

Part 4 — Team Challenge: Raise the roof. Use BOTH rules. Write your plan, then your numbers. Mr. N will give you the ceiling height.

a. The ceiling is _____ cm high. How many FULL rows fit under it?

_____

_____

b. How many cups would that pyramid need?

_____

_____

★ **Think deeper:** Which rule did you need for (a)? For (b)? Why couldn't one rule answer both?

_____

c. The whole class pools 100 cups. What is the TALLEST pyramid you can build? How tall will it be?

rows: _____ cups: _____

predicted height: _____ cm

d. Build it together. Measure it.

actual height: _____ cm

off by: _____ cm

e. Explain why your prediction in (c) was a little off. Name TWO real reasons.

_____

_____

★ **Think deeper:** Forget the pyramid. With the same 100 cups, what is the TALLEST tower you could build? Sketch it and write its height rule.

_____

_____

_____